Assignment-1 Aptitude

1) 25% → 200?

$\frac{25}{100} \times 200 = 50$ (b)

2) 40% $x = 80$ $\frac{40}{100} \times x = 80$ $x = \frac{80 \times 100}{40} = 200$

3) 75% of $x = 150$ $\frac{75}{100} \times x = 150$

$x = \frac{150 \times 100}{75} = 200$

4) 15% → 120? $\frac{15}{100} \times 120 = 18$

5) 30% $x = 90$ $= \frac{30}{100} \times x = 90$, $x = \frac{90 \times 100}{30} = 300$

6) % Increase $= \frac{250-200}{200} \times 100 = 25\%$

7) % Increase $= \frac{50,000 - 40,000}{40,000} \times 100 = 25\%$

8) % ↓ ase $= \frac{10,000 - 8,000}{10,000} \times 100 = 20\%$

9) % ↓ $= \frac{500-400}{500} = 20\%$

10) Loss $= 600 - 450 = 150$

% loss $= \frac{150}{600} \times 100 = 25\%$

(11) 30% 400 $= \frac{30}{100} \times 400 = 120$

40% of 300 $= \frac{40}{100} \times 300 = 120$ } equal

(12) 60% of income = Income − 8,000 ⇒ 0·60 × Income = Incom − 8000
Solving we get 29000.

(13) $A = 1{\cdot}20 \times B \Rightarrow B = \frac{A}{1{\cdot}20} = \frac{100}{120} \times A = 0{\cdot}833 \times A = 16{\cdot}67\%$

(14) New price = 1·25 × old price
New consept $= \frac{\text{old consum}}{1{\cdot}25}$ = 0·8 × old consup.

Reduc. in Consumption = 100% − 80% = 20%

(15) $A = 140 \times B$, $B = \frac{A}{140} = \frac{40}{140} = 0{\cdot}2857 \approx 28.57$

(16) assume old price = 100
New price = 100 × 120 × 0·90 = ₹108
% ket $= \frac{108-100}{100}$ = ~~8%~~ (8% ↑)

(17) assume old = 100
new = 100 × 130 × 0·80 = 104
% Change $= \frac{104-100}{100}$ = (4% ↑)

(18) Assume pop = 100
New pop = 100 × 1·25 × 0·80 = 100
% change = 0

(19) (Old assume = 100) new = 100 × 1·40 × 0·70 = 98
% change $= \frac{98-100}{100}$ = (2% ↓)

(20) let assume $= 100$

New $= 100 + 1.20 + 0.40 = ₹108$

% change $= \frac{108-100}{100} = (8\% \uparrow)$

(21) $SP = CP + 25\%$ of Cost price

$= 1.25 \times CP = 125$

(22) Let CP be x

$SP = 500 \times (1 - 0.10) = 500 \times 0.90 = ₹450$

Profit $= 8\%$ of $CP = 0.08 \times x$

$450 = x + 0.08x \Rightarrow 450 = 1.08x \Rightarrow x = \frac{450}{1.08} = ₹416.67$

(23) Profit $= 20\%$ of $CP = 0.20 \times x$

$SP = x + 0.20 \times x = 1.20 \times x$

profit % $= \frac{0.20x}{1.20x} \times 100 = 16.67\%$

(24) Discount $= 1200 - 960 = ₹240$

% discount $= \frac{240}{1,200} \times 100 = 20\%$

(25) Profit $= 650 - 500 = 150$

% Profit $= \frac{150}{500} \times 100 = 30\%$

(26) $A = 1.20 \times B \Rightarrow B = \frac{A}{1.20}$ $\therefore = \frac{20}{120} = 16.67\%$

(27) Total Student $= 3+2 = 5$

% of Boys $= \frac{3}{5} \times 100 = 60\%$

(28) $\% \uparrow = \frac{250000 - 200000}{200000} \times 100 = \underline{\underline{25}}$

(29) 65% of $V - 35\%$ of $V = 3000 \Rightarrow 0.65V - 0.35V = 3000$

$0.30V = 3000$

$\underline{\underline{V = 10,000}}$

(30) let orignal $= 100$

New $= 100(1 - 0.30) = \underline{70}$

$\% \uparrow = \frac{100 - 70}{70} \times 100 = 42.85\%$

(31) Orignal $= 100$

New $= 100 \times 1.50 \times 0.50 = \underline{75}$

$\%$change $= \frac{75 - 100}{100} \times 100 = 25\% \downarrow$

(32) $B = \frac{100}{120} \times A = 0.833 \times A = 16.67\%$

(33) 30% of $x = 90 \quad x = \frac{90 \times 100}{30} = 300$

60% of $x = 0.60 \times 30 = 180$

(34) 75% of Income $=$ Income $- 5,000 \Rightarrow 0.75 \times$ Income $=$ Income $- 5000$

$0.75 \times \text{Income} - \text{Income} = -5000$

$\text{Income} \times (0.75 - 1) = -5000$

$\text{Income} \times (-0.25) = -5000$

$\text{Income} = \frac{-5000}{-0.25} = 20,000$

(35) New P $= 1.20 \times$ old p

New Con $= \frac{\text{old con}}{1.20} = 0.833 \times$ old con

reduce in con $= 100 - 83.3\% = \boxed{16.67\%}$

(36) Orignal price = 100
MP = $100 \times 1.25 = ₹125$
SP = $125 \times (1 - 0.20) = 100$
P/loss % = 0%

(38) Loss 20% of 500 = 100
SP = 500 − 100 = ₹400

(39) Orignal Price = 100
New Sal = $100 \times 1.10 \times 0.90 = ₹99$
Net % Change = $\frac{99 - 100}{100} \times 100 = 1\%$ decrease

(40) PMS = 200 + 20 = 220
40% of Total Marks = 220 ⇒ $\frac{40}{100} \times$ Total Marks = 220

Total Marks = 550

(41) Total expense 20% + 30% + 10% = 60%
Savings = 40% of salary ⇒ 0.40 × Salary = 18,000
⇒ Salary = 45000

(42) NP = $100 \times 1.30 \times 0.70 = 91$
% Change = $\frac{91 - 100}{100} \times 100 =$ (9% ↓)

(43) Future Pop = $10000 \times 1.10^3 = 13,310$

(44) 0.15A = 0.20B ⇒ A:B = 4:3

(45) SP = $800 * (1 + 0.25) = 1000$

(46) Profit = 250 − 200 = ₹50

$\% \text{profit} = \frac{50}{200} \times 100 = 25\%$

(47) $SP = 720 = CP \times (1 + 0.20) \Rightarrow ₹0$

$720 = 1.20 \times CP$

$CP = \frac{720}{1.20} = ₹600$

(48) $MP = 75\%$ of OP

$\% f = \frac{100-75}{75} \times 100 = 33.33\%$

(49) Sp on Res = $0.40 \times 15{,}000 = ₹6{,}000$

Sp on food = $0.30 \times 15{,}000 = ₹4{,}500$

T. Spending :- $6{,}000 + 4500 = 10{,}500$

Save = $15000 - 10500 = \underline{\underline{4{,}500}}$

(50) $SP = 400 \times (1 + 0.25) = \underline{\underline{500}}$